

HWK 1
ADVANCED MACHINE LEARNING
DATA 642

Objective: The homework objectives encompass various mathematical and optimization concepts. Firstly, students will determine the minimum of a cubic function and analyze the positive definiteness of the Hessian matrix at a specific point. They will then investigate the convergence behavior of steepest descent algorithms and prove properties related to symmetric positive matrices. Next, they'll demonstrate convexity using second derivative analysis and apply gradient descent to optimize a given objective function. Additionally, students will implement a gradient descent algorithm for function optimization, observing its behavior with different initialization points. Finally, they'll apply Newton's method and gradient descent to optimize a specified loss function, comparing their convergence behavior in terms of the number of iterations.

Exercise 1 (1 point)

What is the minimum of the function $J(\theta) = \theta^3 - \theta$?

Exercise 2 (1 point)

Let $J(\boldsymbol{\theta}) = (1 - \theta_1)^2 + 5(\theta_2 - \theta_1^2)^2$. Is the Hessian positive definite at the point $[1, 1]$?

Exercise 3 (1 point)

Do all steepest descent algorithms lead to the same model provided you let them run long enough? Justify your answer.

Exercise 4 (1 point)

Show that the eigenvalues of a symmetric positive matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ are all positive. (Hint: Recall that the eigenvalues of a symmetric matrix are real.)

Exercise 5 (1 point)

Show that the negative entropy function $J(\theta) = \theta \log \theta$ is convex for all $\theta > 0$. Here $J : \mathbb{R} \rightarrow \mathbb{R}$. (Hint: If we know that a function is twice differentiable, that is, the Hessian exists for all values in the domain of θ , then the function is convex if and only if the Hessian is positive semi-definite.)

Exercise 6 (2 points)

Show that the objective function $J(\boldsymbol{\theta}) = \frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|^2$ is convex for any invertible matrix $\mathbf{X}$. Find global optimum by computing the gradient, setting it equal to zero, and solving for $\boldsymbol{\theta}$.

Exercise 7 (3 points) Consider the function $J(\boldsymbol{\theta}) = \theta_1^2 + \theta_2^{10}$. What is the optimal value? Implement a simple gradient descent algorithm with a constant step-size/learning rate. Consider two different initialization points $[0.5, 0.5]$ and $[2, 2]$ and plot the value of J as a function of the number of iterations. What do you observe about the behavior of the algorithm in the two cases?

Exercise 8 (5 points) You are a data scientist working for a cutting-edge artificial intelligence research company, AI Innovations Inc. Your team is currently working on developing a machine

learning model to improve the accuracy of medical image classification for diagnosing various diseases. The model architecture is complex, and training it involves optimizing the loss function $J(\boldsymbol{\theta}) = \frac{1}{2}\boldsymbol{\theta}^\top \mathbf{H}\boldsymbol{\theta}$. Your task is to implement a second-order optimization method, specifically Newton's method, to improve the training efficiency and convergence.

Apply Newton's method to $J(\boldsymbol{\theta}) = \frac{1}{2}\boldsymbol{\theta}^\top \mathbf{H}\boldsymbol{\theta}$ starting from a random point of $\boldsymbol{\theta}$. Use

$$\mathbf{H} = \begin{bmatrix} 1 & 0 \\ 0 & 1000 \end{bmatrix} \quad (1)$$

What do you observe in terms of the numbers of iterations needed for the algorithm to converge? Next apply gradient descent to the same optimization problem. Which algorithm is faster in terms of number of iterations needed to convergence.

Hint: For the implementation and details of Newton method use the following link https://xavierbourretsicotte.github.io/Intro_optimization.html#Newton-Raphson's-method